

CIFAR-10 IMAGE CLASSIFICATION

Presented by:

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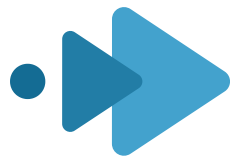
THE IMAGES

Each CIFAR-10 image is just 32×32 pixels, smaller than the app icons on your phone. Across the dataset there are 10 everyday categories:

- planes
- cars
- birds
- cats
- deer
- dogs
- frogs
- horses
- ships
- trucks



Can a model still tell that this is a **ship**?



WHY IT MATTERS



- Solving image classification at low resolution makes it possible at large scale.
- The same techniques power medical imaging, satellite analysis, accessibility tools, and edge devices where compute and bandwidth are limited.



THE DATASET



- The full CIFAR-10 dataset contains 60,000 colour images, evenly split across the ten categories.
- To keep training time manageable, I worked with a 10,000-image training subset and held back the full 10,000-image test set for final evaluation.

Sample Training Images per Class

airplane



automobile



bird



cat



deer



dog



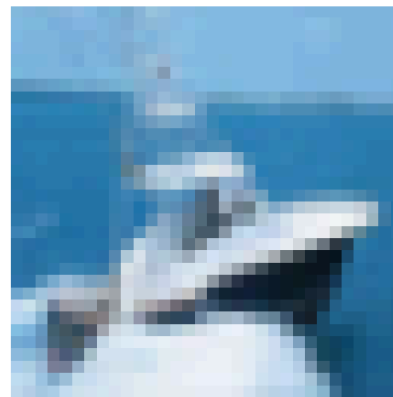
frog



horse



ship



truck



PREPROCESSING



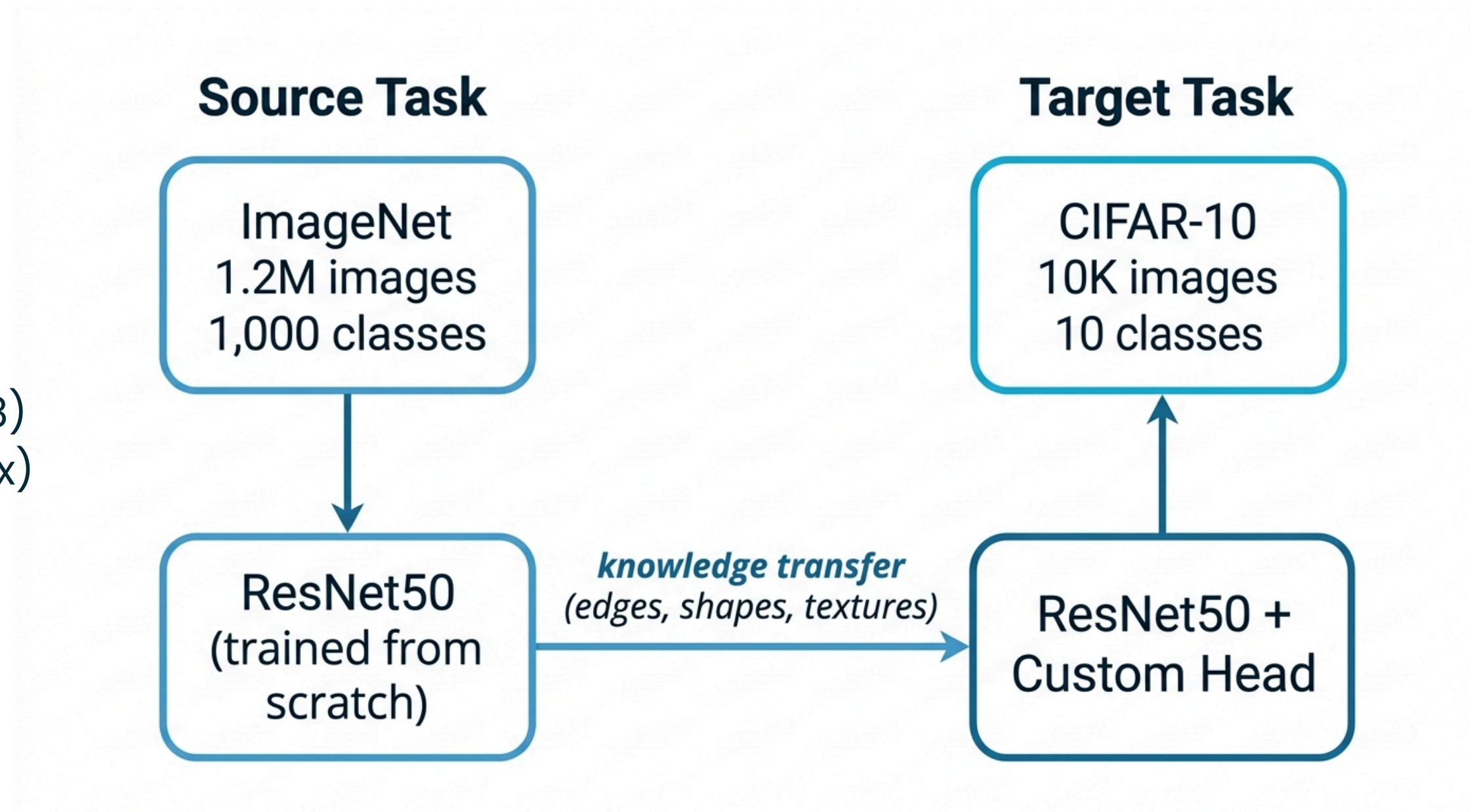
- Pixel values normalised from 0–255 to 0–1 for stable training
- One-hot encoded labels for 10-class output
- Held-out test set untouched until final evaluation

Humble input, ambitious output



TRANSFER LEARNING

- Why train from scratch when ImageNet has already learned shapes, edges, and textures from 1.2 million images?
- Used ResNet50 pre-trained on ImageNet as the backbone (top classification layer removed)
- Added a custom classification head: GlobalAvgPool → Dense(128) → Dense(64) → Dense(10, Softmax)



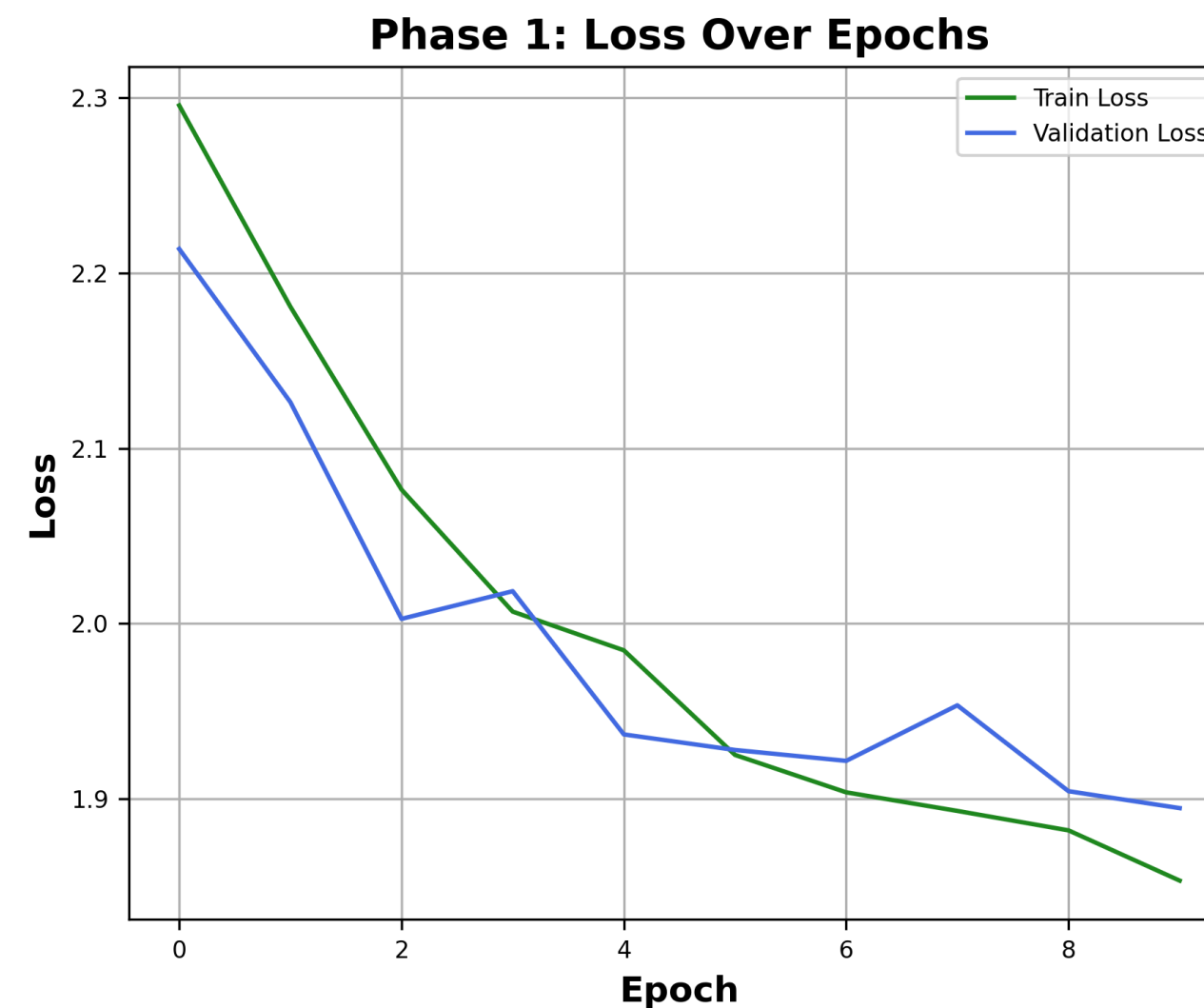
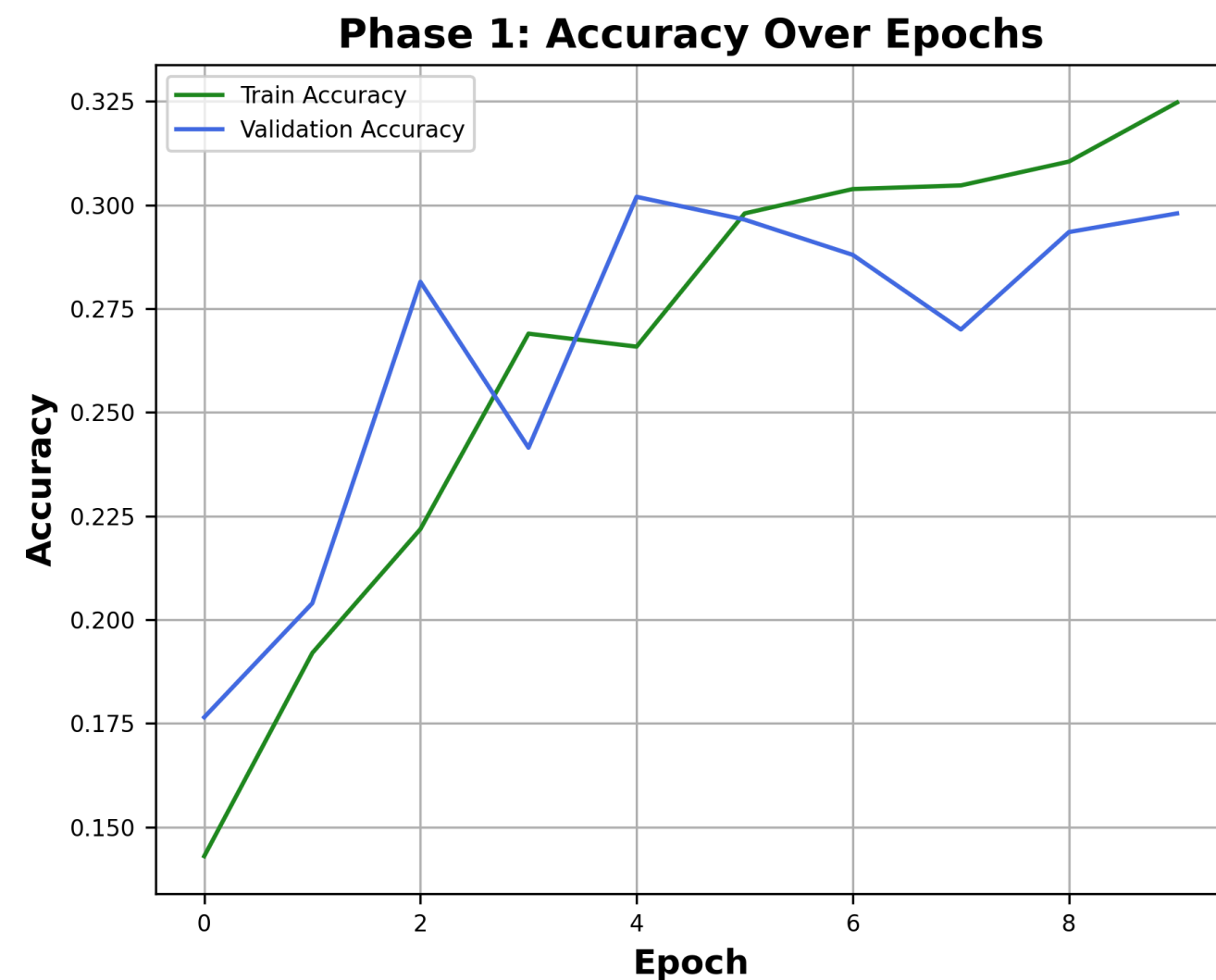
PHASE 1: FROZEN BASE



Strategy: Froze ResNet50, trained only the custom head for 10 epochs

Why: Protect ImageNet knowledge, let the head learn the new classes

Takeaway: ~30% is the ceiling of frozen ImageNet features on these tiny images.
To do better, we need to let the base adapt.



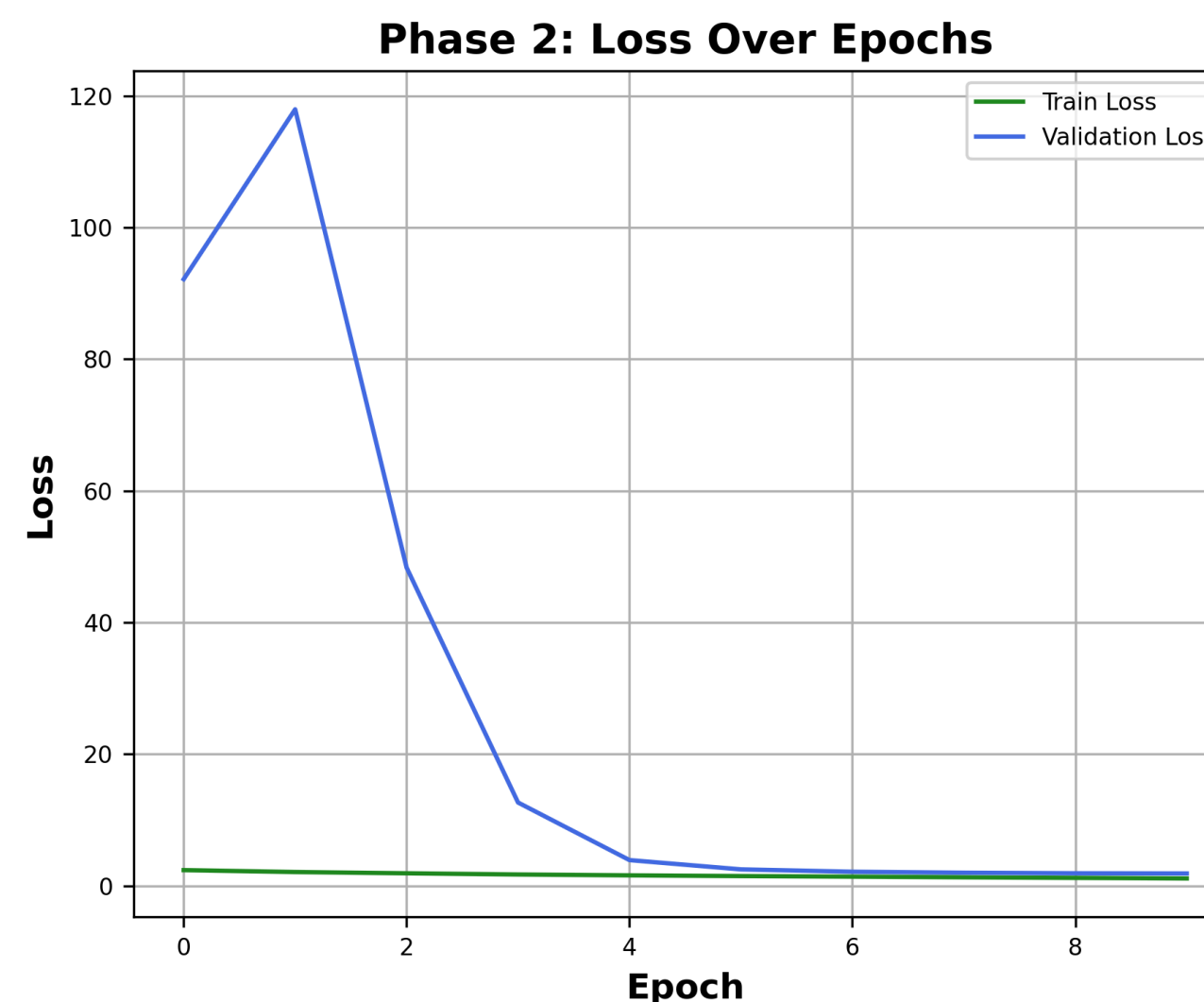
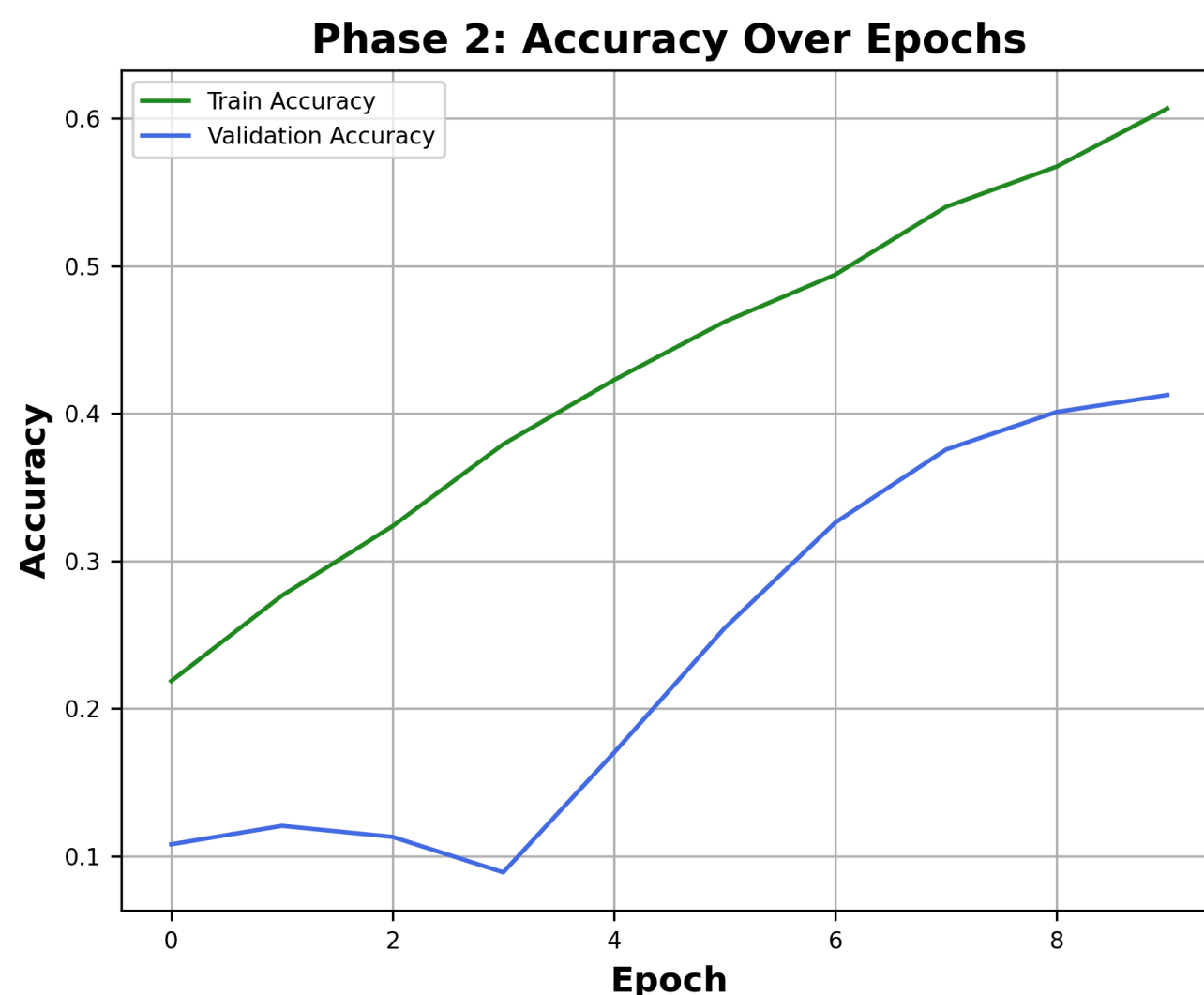
PHASE 2: FINE-TUNING



Strategy: Unfroze the entire model, continued training for 10 more epochs at a tiny learning rate ($1e-5$)

Why: Let ResNet50 gently adapt its features to CIFAR's small images without forgetting ImageNet

Takeaway: Training accuracy doubled to ~60%, validation reached ~41%. The gap signals overfitting, and the early validation loss spike shows the model destabilising as it readjusts to all weights being trainable.



FINAL EVALUATION



Test accuracy: 40.2%

Baseline (random guessing): 10%

Result: 4× better than chance

Takeaway: A clear diagonal forms in the confusion matrix, the model has genuinely learned to discriminate between classes. But some classes shine, others struggle, and the off-diagonal patterns reveal which classes the model confuses with each other.

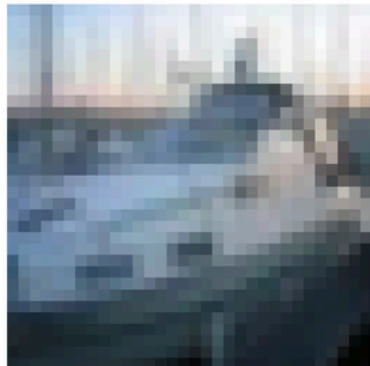
Confusion Matrix

True Label	airplane	automobile	bird	cat	deer	dog	frog	horse	ship	truck
	445	50	102	35	43	10	20	33	207	55
	87	444	11	30	13	17	59	47	92	200
	109	26	285	83	152	80	133	75	30	27
	23	43	87	302	72	153	146	82	37	55
	50	14	113	71	373	57	167	99	30	26
	16	28	91	217	97	303	112	83	8	45
	30	40	56	85	138	59	496	46	23	27
	32	36	80	103	106	82	57	413	14	77
	194	80	35	28	18	7	26	15	522	75
Predicted Label	airplane	automobile	bird	cat	deer	dog	frog	horse	ship	truck
	66	196	19	42	11	16	31	83	95	441

WHAT THE MODEL GOT RIGHT

Correct Predictions

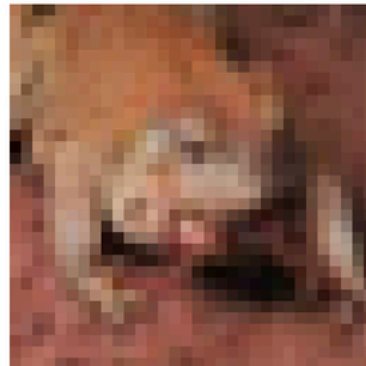
Predicted: ship
True: ship



Predicted: airplane
True: airplane



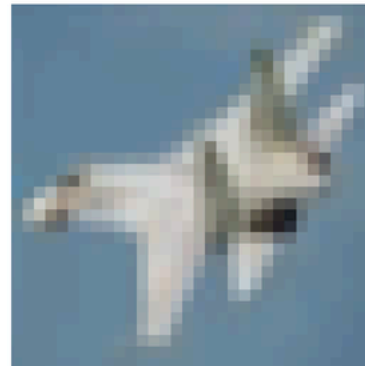
Predicted: frog
True: frog



Predicted: automobile
True: automobile



Predicted: airplane
True: airplane



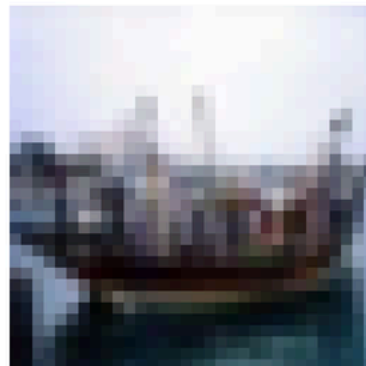
Predicted: truck
True: truck



Predicted: truck
True: truck



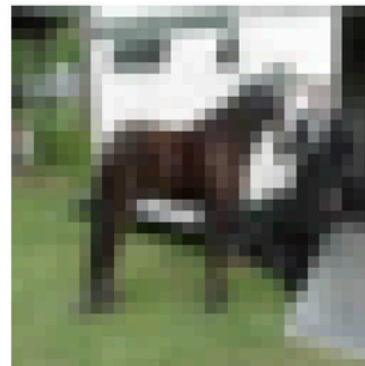
Predicted: ship
True: ship



Predicted: frog
True: frog



Predicted: horse
True: horse



Predicted: bird
True: bird



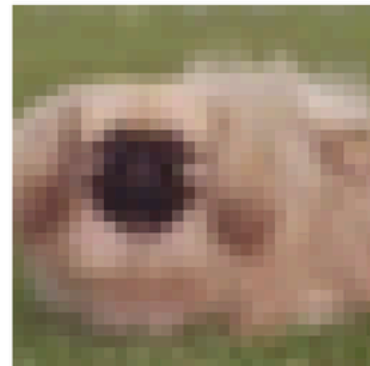
Predicted: truck
True: truck



Predicted: frog
True: frog



Predicted: dog
True: dog



Predicted: deer
True: deer



Top performers (F1 score):

- Ship: 51%
- Automobile: 45%
- Frog: 44%

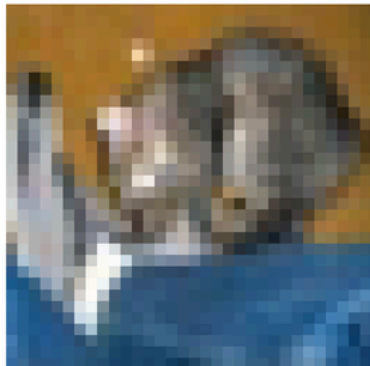
Why:

- Distinctive shapes
- Strong colour signatures
- Consistent backgrounds

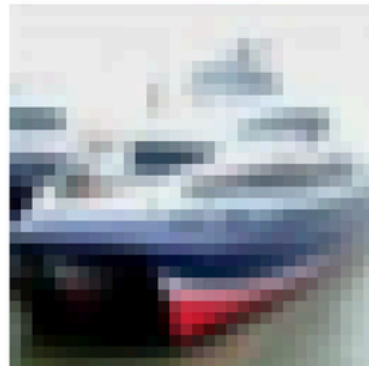
WHAT THE MODEL GOT WRONG

Wrong Predictions

Predicted: frog
True: cat



Predicted: automobile
True: ship



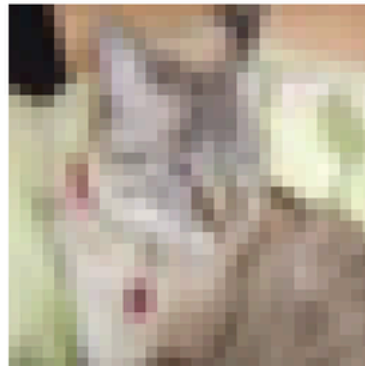
Predicted: deer
True: frog



Predicted: dog
True: frog



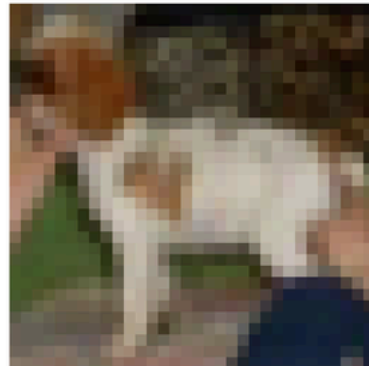
Predicted: dog
True: cat



Predicted: truck
True: automobile



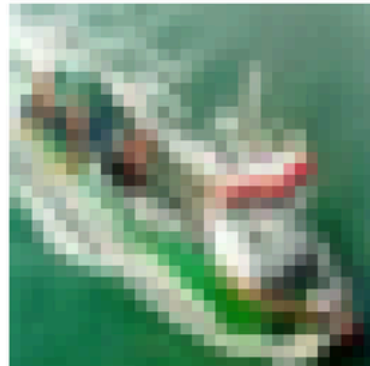
Predicted: frog
True: dog



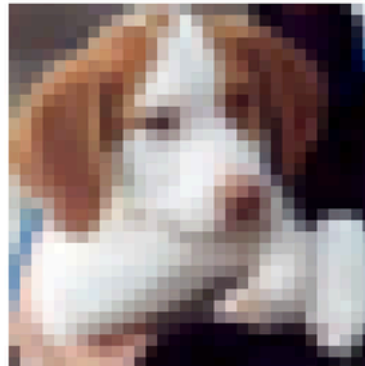
Predicted: frog
True: horse



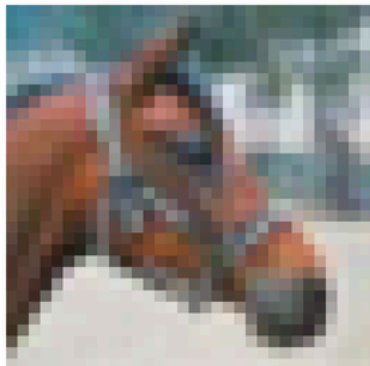
Predicted: airplane
True: ship



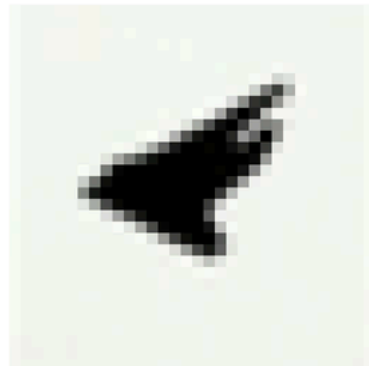
Predicted: truck
True: dog



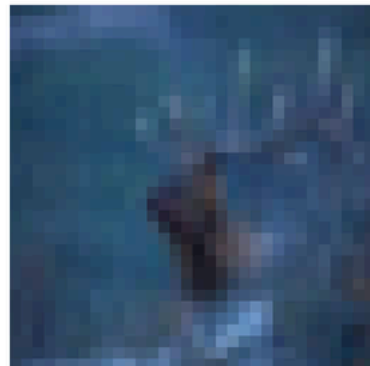
Predicted: deer
True: horse



Predicted: deer
True: airplane



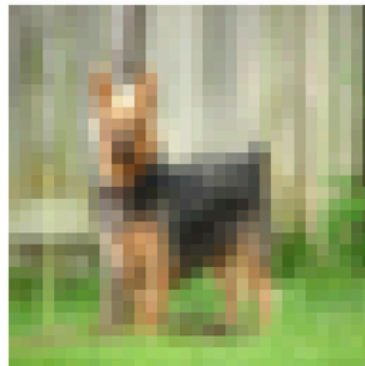
Predicted: dog
True: deer



Predicted: automobile
True: truck



Predicted: cat
True: dog



Weakest performers (F1 score):

- Bird: 30%
- Cat: 30%
- Dog: 34%

Common confusions:

- Cat ↔ Dog
- Automobile ↔ Truck
- Airplane ↔ Ship

Why:

- Visually similar silhouettes
- Variable poses and angles
- 32×32 resolution loses fine detail

LIMITATIONS & IMPROVEMENTS

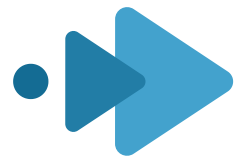


Limitations

- Trained on only 10K of 50K available images
- No data augmentation applied
- ~19% train-validation gap = overfitting
- 32×32 resolution itself loses fine-grained detail
- ResNet50 is heavy for such small inputs

Future Improvements

- Data augmentation (flips, rotations, colour jitter)
- Train on the full 50K set
- Add dropout regularisation to combat overfitting
- Use early stopping based on validation loss
- Try lighter architectures (EfficientNet-B0, MobileNetV2)



THANK YOU FOR WATCHING!

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